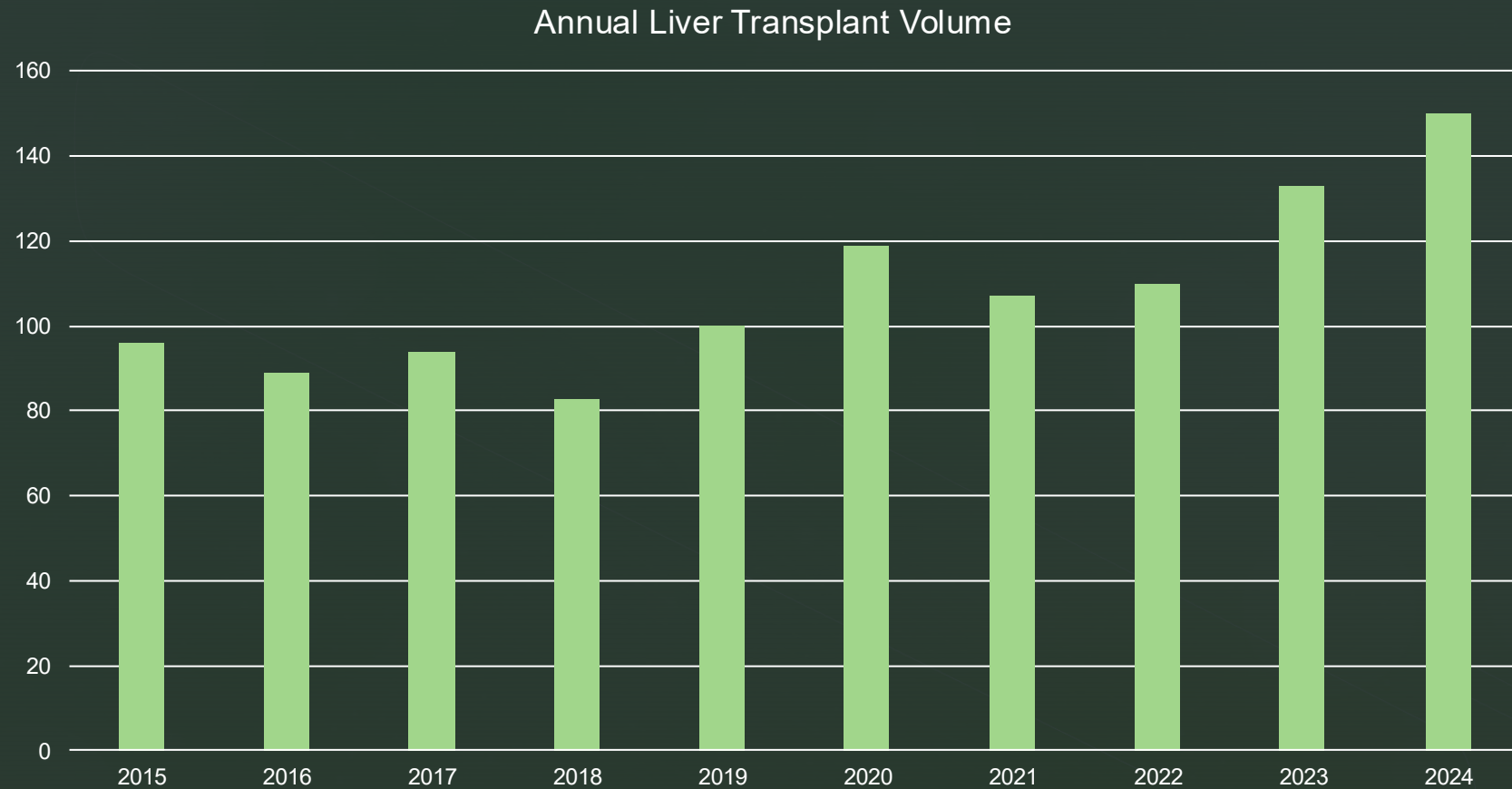


Faruq Pradhan MBBCh, FRCPC

Nutritional Success & Liver Transplantation



10-Year UNMC Transplant Volume



Center data - OPTN. (n.d.). Retrieved February 16, 2025, from <https://optn.transplant.hrsa.gov/data/view-data-reports/center-data/#>

Nutritional Deficiencies and Cirrhosis

- 60-70% of patients with decompensated cirrhosis are malnourished
- Micronutrient deficiencies are common, especially in those with chronic cholestasis



Mishra, S., & Premkumar, M. (2023). Nutritional management of a liver transplant candidate.

Journal of Clinical and Experimental Hepatology, 13(5), 878–894.

Intervention

- Routinely screen all pre-operative patients presenting to hospital for liver transplant surgery for the following deficiencies:
 - Vitamin A
 - Vitamin C
 - Vitamin D
 - Vitamin E
 - Zinc
 - Copper
- If present, initiate repletion with PO or IV supplementation as necessary and monitor levels



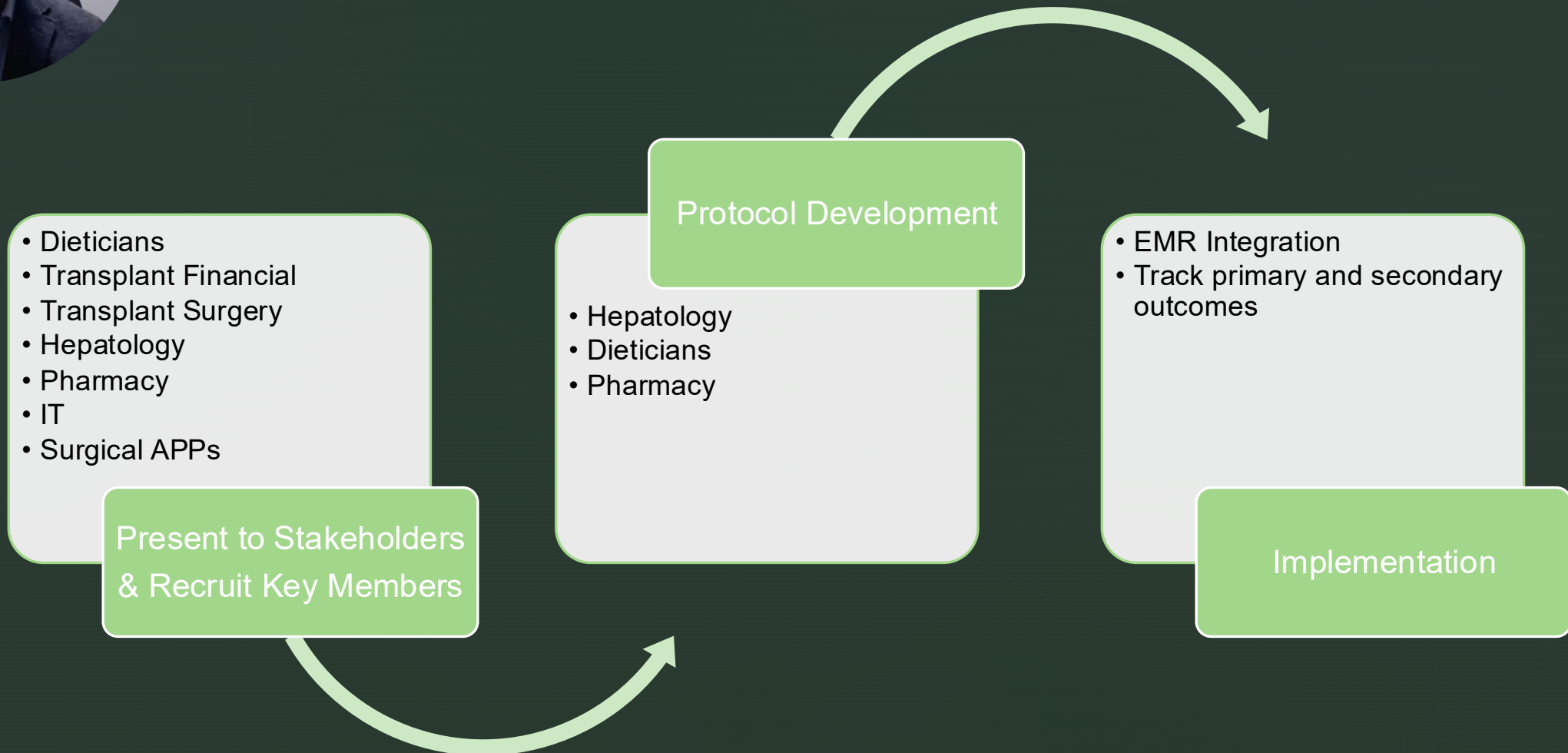
Outcomes

- Primary outcome:
 - Length of stay post-transplant
- Secondary outcomes:
 - 30-day wound complications
 - Unexpected return to OR during index admission
 - Re-hospitalization rate over 90 days
 - Rate of infections (culture-positive) over 90 days
 - Acute cellular rejection incidence





Pathway to Implementation



Overall Project Needs

- Support from Transplant Surgery and Transplant Hepatologists
- 'Champions' in the various disciplines to aid with implementation
- Hepatology fellow for data review, abstract and manuscript publication



Tasks Per Role: Dietician

- Protocol creation for micronutrient replacement, including dosing
 - Work with pharmacy on appropriate dosing to be included in protocol
- Review with UNMC lab how long they expect results after labs are drawn
- Establish monitoring intervals for serial lab assessment of deficient vitamins



Tasks Per Role: Pharmacy

- Ensure oral and IV vitamin replacements are available on formulary
- Work in tandem with Dieticians and myself to create replacement protocol
- Identify when to check follow-up levels



Tasks Per Role: Hepatology Fellow

- 6-9 months post-intervention:
 - Review project with Biostatistician to plan retrospective data collection of outcomes
 - Institutional Review Board application
 - Present analyzed data back to group



Tasks Per Role: Surgical APP

- Discuss vitamin/micronutrient labs during multi-disciplinary rounds
- Ensure supplements are continued at discharge



Potential Barriers & Opportunities

- Lab capacity to run labs
- Potential downstream cost to patients
- No current established guidelines



Outcome Review & Future Goals

- Key data outcomes reviews post-implementation
 - At 6-9 months
 - At 12-15 months
- Future:
 - Expand other solid-organ transplant groups
 - Sub-group analysis in multi-organ transplants that include liver



Questions?

